

Register Transfer and Microoperations

AMAN DEEP SINGH
ASSISTANT PROFESSOR
GLA UNIVERSITY MATHURA

SYLLABUS

- ▶ Register Transfer Language
- ▶ Bus and Memory Transfers
- ▶ Arithmetic
- ▶ Logic and Shift Microoperations

Register Transfer Language

- ▶ In symbolic notation, it is used to describe the micro-operations transfer among registers.
- ▶ It is a kind of intermediate representation (IR) that is very close to assembly language, such as that which is used in a compiler.
- ▶ The term “Register Transfer” can perform micro-operations and transfer the result of operation to the same or other register.
- ▶ **Micro-operations :**
The operation executed on the data store in registers are called micro-operations. They are detailed low-level instructions used in some designs to implement complex machine instructions.
- ▶ **Register Transfer :**
The information transformed from one register to another register is represented in symbolic form by replacement operator is called Register Transfer.

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► Replacement Operator :

In the statement, $R2 \leftarrow R1$, $\leftarrow$ acts as a replacement operator. This statement defines the transfer of content of register R1 into register R2.

Letters and Numbers	Denotes a Register	MAR, R1, R2
()	Denotes a part of register	R1(8-bit) R1(0-7)
$\leftarrow$	Denotes a transfer of information	$R2 \leftarrow R1$
,	Specify two micro-operations of Register Transfer	$R1 \leftarrow R2$ $R2 \leftarrow R1$

Register Transfer Operations:

- ▶ The operation performed on the data stored in the registers are referred to as register transfer operations.
- ▶ There are different types of register transfer operations:
- ▶ **1. Simple Transfer - $R2 \leftarrow R1$**
- ▶ The content of R1 are copied into R2 without affecting the content of R1. It is an unconditional type of transfer operation.
- ▶ **2. Conditional Transfer -**
- ▶ It indicates that if $P=1$, then the content of R1 is transferred to R2. It is a unidirectional operation.
- ▶ **3. Simultaneous Operations -**
- ▶ If 2 or more operations are to occur simultaneously then they are separated with comma (,).

Bus and Memory Transfers

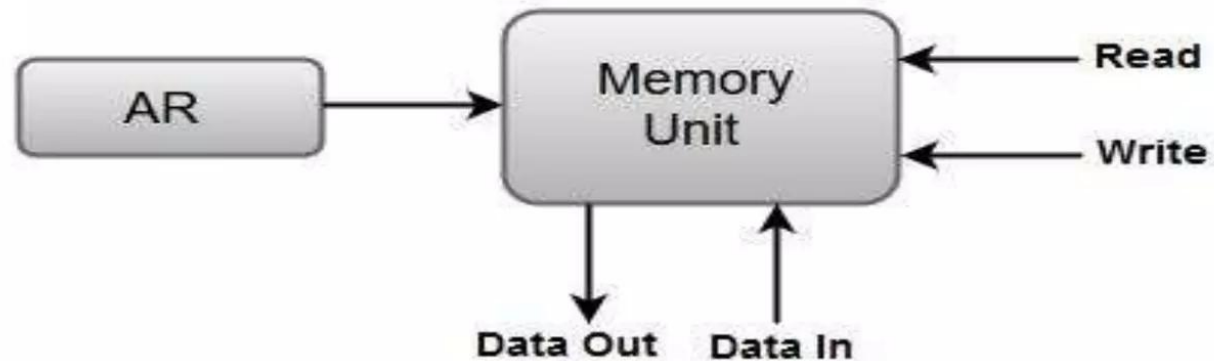
- ▶ A digital system composed of many registers, and paths must be provided to transfer information from one register to another.
- ▶ The number of wires connecting all of the registers will be excessive if separate lines are used between each register and all other registers in the system.
- ▶ A bus structure, on the other hand, is more efficient for transferring information between registers in a multi-register configuration system.
- ▶ A bus consists of a set of common lines, one for each bit of register, through which binary information is transferred one at a time.
- ▶ Control signals determine which register is selected by the bus during a particular register transfer.

Memory Transfer

- ▶ Most of the standard notations used for specifying operations on memory transfer are stated below.
- ▶ The transfer of information from a memory unit to the user end is called a **Read** operation.
- ▶ The transfer of new information to be stored in the memory is called a **Write** operation.
- ▶ A memory word is designated by the letter **M**.
- ▶ We must specify the address of memory word while writing the memory transfer operations.
- ▶ The **address register** is designated by **AR** and the **data register** by **DR**.
- ▶ Thus, a read operation can be stated as:
 - ▶ Read: $DR \leftarrow M[AR]$
- ▶ The **Read** statement causes a transfer of information into the data register (DR) from the memory word (M) selected by the address register (AR).

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- ▶ And the corresponding write operation can be stated as:
- ▶ Write: $M[AR] \leftarrow R1$
- ▶ The Write statement causes a transfer of information from register R1 into the memory word (M) selected by address register (AR).



Arithmetic Logic and Shift Microoperations

- ▶ In general, the Arithmetic Micro-operations deals with the operations performed on numeric data stored in the registers.
- ▶ The basic Arithmetic Micro-operations are classified in the following categories:
 - ▶ Addition
 - ▶ Subtraction
 - ▶ Increment
 - ▶ Decrement
 - ▶ Shift

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- ▶ Some additional Arithmetic Micro-operations are classified as:
- ▶ Add with carry
- ▶ Subtract with borrow
- ▶ Transfer/Load, etc.

Symbolic Representation	Description
$R3 \leftarrow R1 + R2$	The contents of R1 plus R2 are transferred to R3.
$R3 \leftarrow R1 - R2$	The contents of R1 minus R2 are transferred to R3.
$R2 \leftarrow R2'$	Complement the contents of R2 (1's complement)
$R2 \leftarrow R2' + 1$	2's complement the contents of R2 (negate)
$R3 \leftarrow R1 + R2' + 1$	R1 plus the 2's complement of R2 (subtraction)
$R1 \leftarrow R1 + 1$	Increment the contents of R1 by one
$R1 \leftarrow R1 - 1$	Decrement the contents of R1 by one

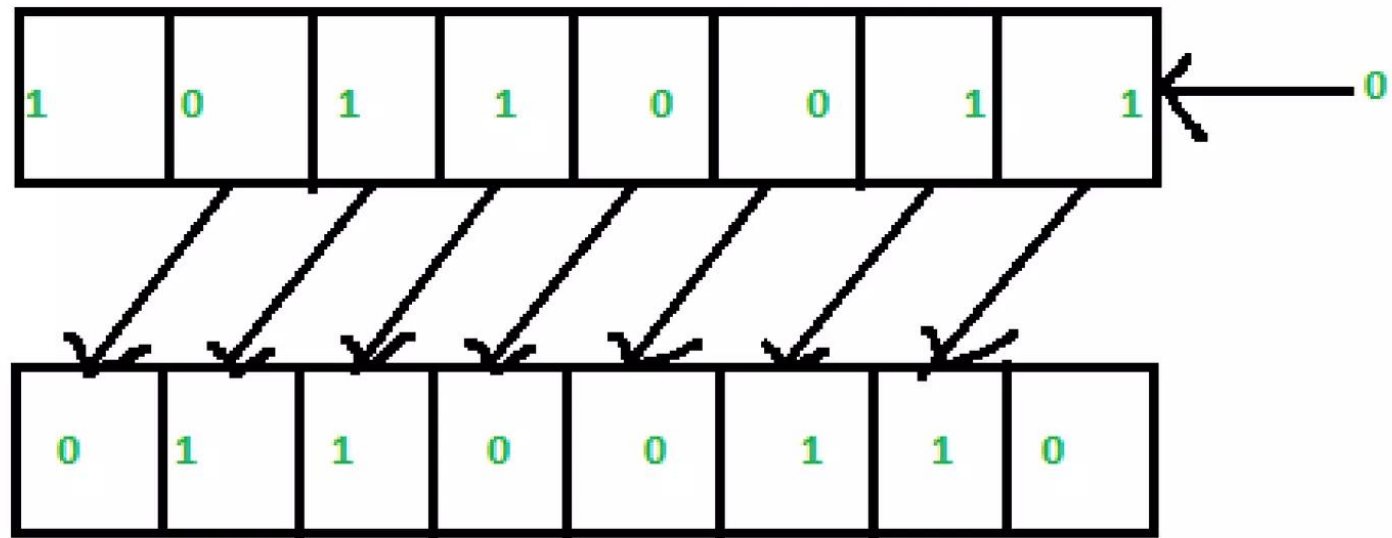
Shift micro-operations

- ▶ Shift micro-operations are those micro-operations that are used for serial transfer of information. These are also used in conjunction with arithmetic micro-operation, logic micro-operation, and other data-processing operations.
- ▶ There are three types of shifts micro-operations:
- ▶ **1. Logical :**
It transfers the 0 zero through the serial input. We use the symbols shl for logical shift-left and shr for shift-right.
- ▶ **Logical Shift Left -**
In this shift one position moves each bit to the left one by one. The Empty least significant bit (LSB) is filled with zero (i.e, the serial input), and the most significant bit (MSB) is rejected.

Left Logical Shift

MSB

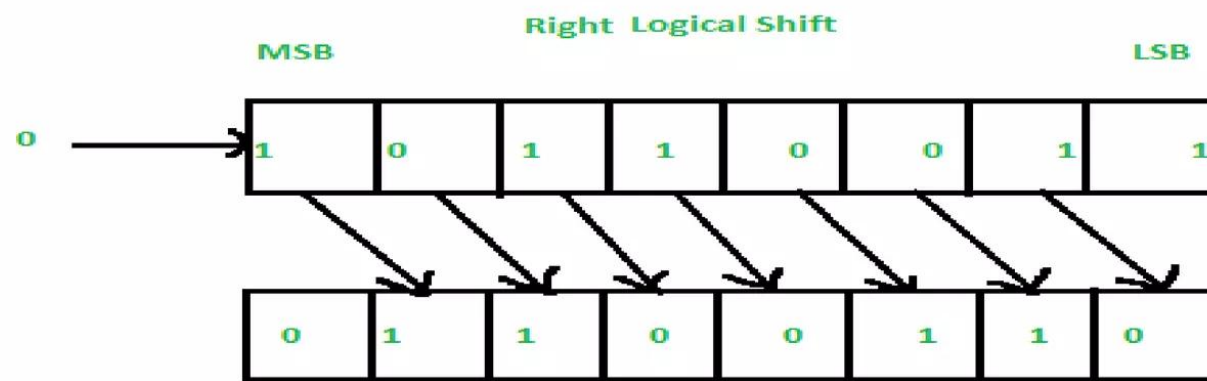
LSB



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► Right Logical Shift -

In this one position moves each bit to the right one by one and the least significant bit(LSB) is rejected and the empty MSB is filled with zero.



Thank you😊

